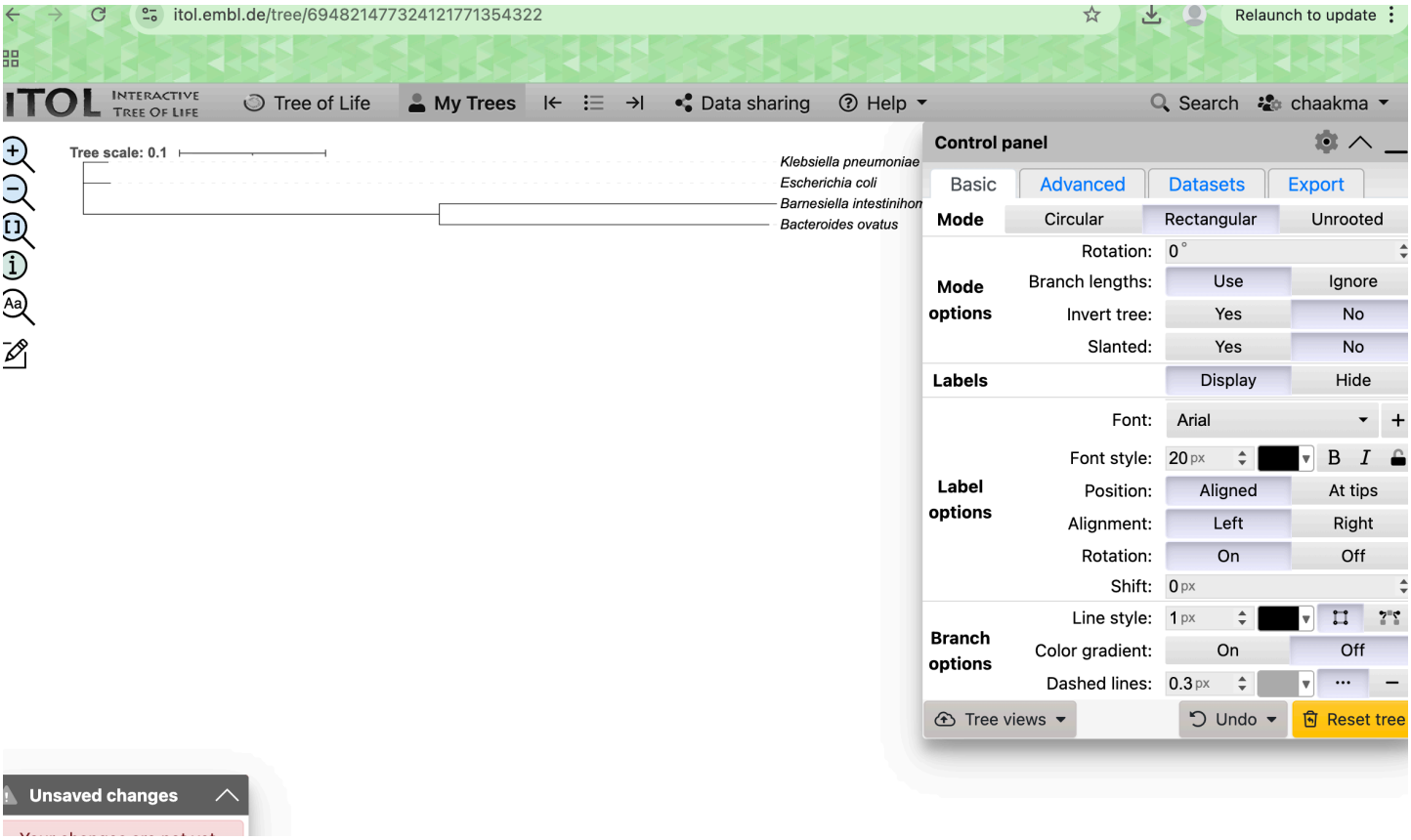
BIOL 4810 — Day 7 Submission

Part 1 — FastME (distance-based)

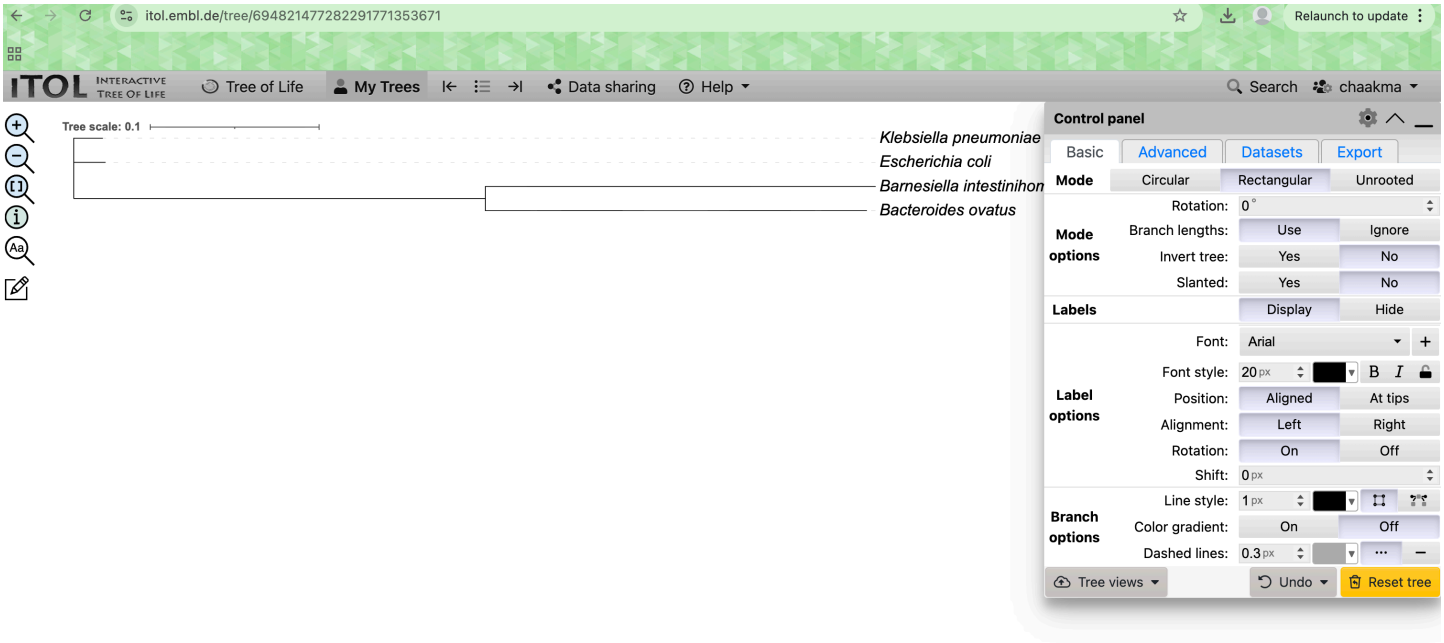
A1: The DNA and protein trees do have the same topology. The species have the same most recent common ancestors in both trees.

A2: The neighbor joining did not do the bootstrap; the previous website did not work.

FastME DNA tree:



FastME Protein tree:



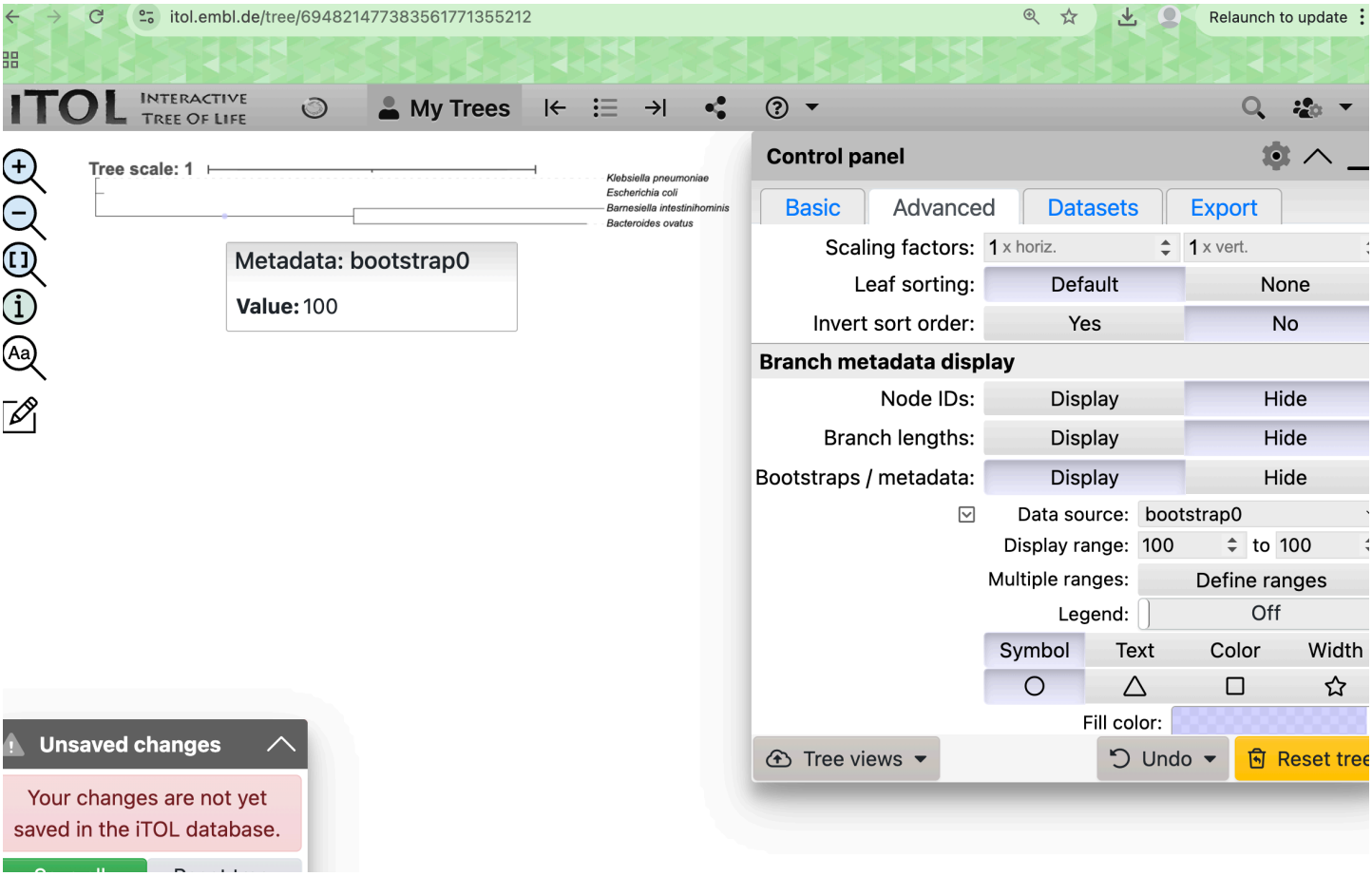
Part 2 — PhyML-SMS (maximum likelihood)

B1: I built the phylogenetic tree utilizing the other website.

B2: There is no difference between the branches of the DNA and the protein tree.

B3: There were no bootstrap values for the first phylogenetic tree, but my bootstrap value for the second tree was 100, which indicates maximum confidence in the grouping of the taxa.

ML Protein tree (PhyML-SMS):



Part 3 — Rooting (concepts)

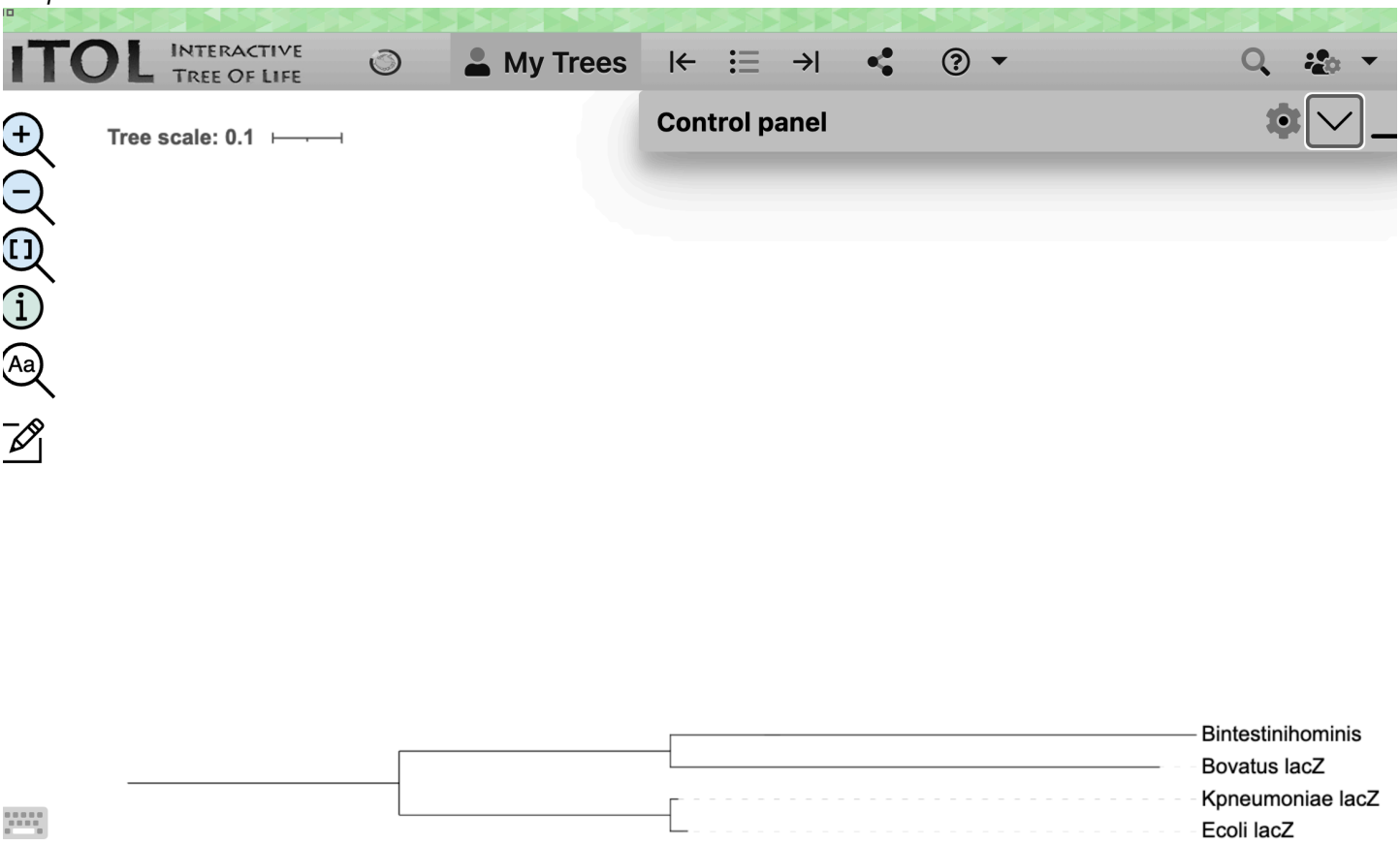
C1: The midpoint rooting changes the interpretation because it demonstrates more than the relationships between taxa. Midpoint rooting allows for interpretation of ancestral and derived character states. It assumes that the two most distant points have evolved at relatively the same rate, hence assuming a molecular clock.

C2: The effect of midpoint rooting on the ML protein tree is similar to the other protein tree because it allows for the interpretation of derived and ancestral character states.

C3: Midpoint reading makes the assumption that the phylogenetic tree follows a molecular clock, meaning that the most divergent taxa are evolving at the same rate.

Part 4 — Synthesis + rooted ML

Midpoint-rooted ML Protein tree:



D1: Rooting is the comparison that changed my interpretation the most. The phylogenetic trees for the protein exhibited different compositions when they were unrooted vs. midpoint rooted. Rooting helped to change my direction of evolution. Rooting changed the composition of my tree the most, and demonstrated both the ancestral and derived character states.

D2: This activity reinforces the concept that phylogenetic trees are supported models of evolutionary change, as opposed to an unchangeable fact of evolution. When new molecular or morphological evidence comes about, then the phylogenetic tree can be changed to best support evolutionary lineages.